

PART A: Input, Output & Conditions (10 Questions)

Q1. Write a program that takes an integer input and prints whether it is positive, negative, or zero.

Q2. Write a program that takes age as input and prints:

- “Eligible for job” if $\text{age} \geq 21$
- Otherwise print “Not eligible”

Q3. Write a program that takes marks of three subjects and prints the average and grade.

Q4. Write a program that takes a number and checks whether it is divisible by both 3 and 5.

Q5. Write a program that asks the user for a character and checks whether it is an alphabet, digit, or special character.

Q6. Write a program to input a year and check whether it is a leap year or not.

Q7. Write a program that takes temperature as input and prints:

- “Very Hot” (>40)
- “Hot” (30–40)
- “Normal” (15–29)
- “Cold” (<15)

Q8. Write a program to take three numbers and print them in ascending order.

Q9. Write a program that takes salary as input and calculates bonus:

- 10% if $\text{salary} \geq 50000$
- 5% otherwise

Q10. Write a program to check whether a given number is a valid 3-digit number.

◇ PART B: Loops & Number Logic (10 Questions)

Q11. Write a program to print all even numbers between 1 and N.

Q12. Write a program to calculate the factorial of a number using a loop.

Q13. Write a program to reverse a given number.

Q14. Write a program to check whether a number is a palindrome.



Q15. Write a program to print the Fibonacci series up to N terms.

Q16. Write a program to count the number of digits in a given number.

Q17. Write a program to find the sum of digits of a number.

Q18. Write a program to check whether a number is an Armstrong number.

Q19. Write a program to print all prime numbers between 1 and N.

Q20. Write a program to find the greatest common divisor (GCD) of two numbers.

◇ PART C: Strings (8 Questions)

Q21. Write a program to count the number of vowels in a string.

Q22. Write a program to reverse a string without using built-in reverse functions.

Q23. Write a program to check whether a string is a palindrome.

Q24. Write a program to count the frequency of each character in a string.

Q25. Write a program to remove all spaces from a given string.

Q26. Write a program to find the longest word in a sentence.

Q27. Write a program to replace all vowels in a string with *.

Q28. Write a program to check whether two strings are anagrams.

◇ PART D: Lists, Tuples & Dictionaries (9 Questions)

Q29. Write a program to find the largest and smallest element in a list.

Q30. Write a program to remove duplicate elements from a list.

Q31. Write a program to find the second largest number in a list.

Q32. Write a program to count even and odd numbers in a list.

Q33. Write a program to merge two lists into one list.

Q34. Write a program to sort a list without using the `sort()` method.

Q35. Write a program to create a dictionary of numbers and their squares from 1 to N.



Q36. Write a program to find the key with the maximum value in a dictionary.

Q37. Write a program to convert a list into a tuple and a tuple into a list.

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